Optical telescopes

Telescopes

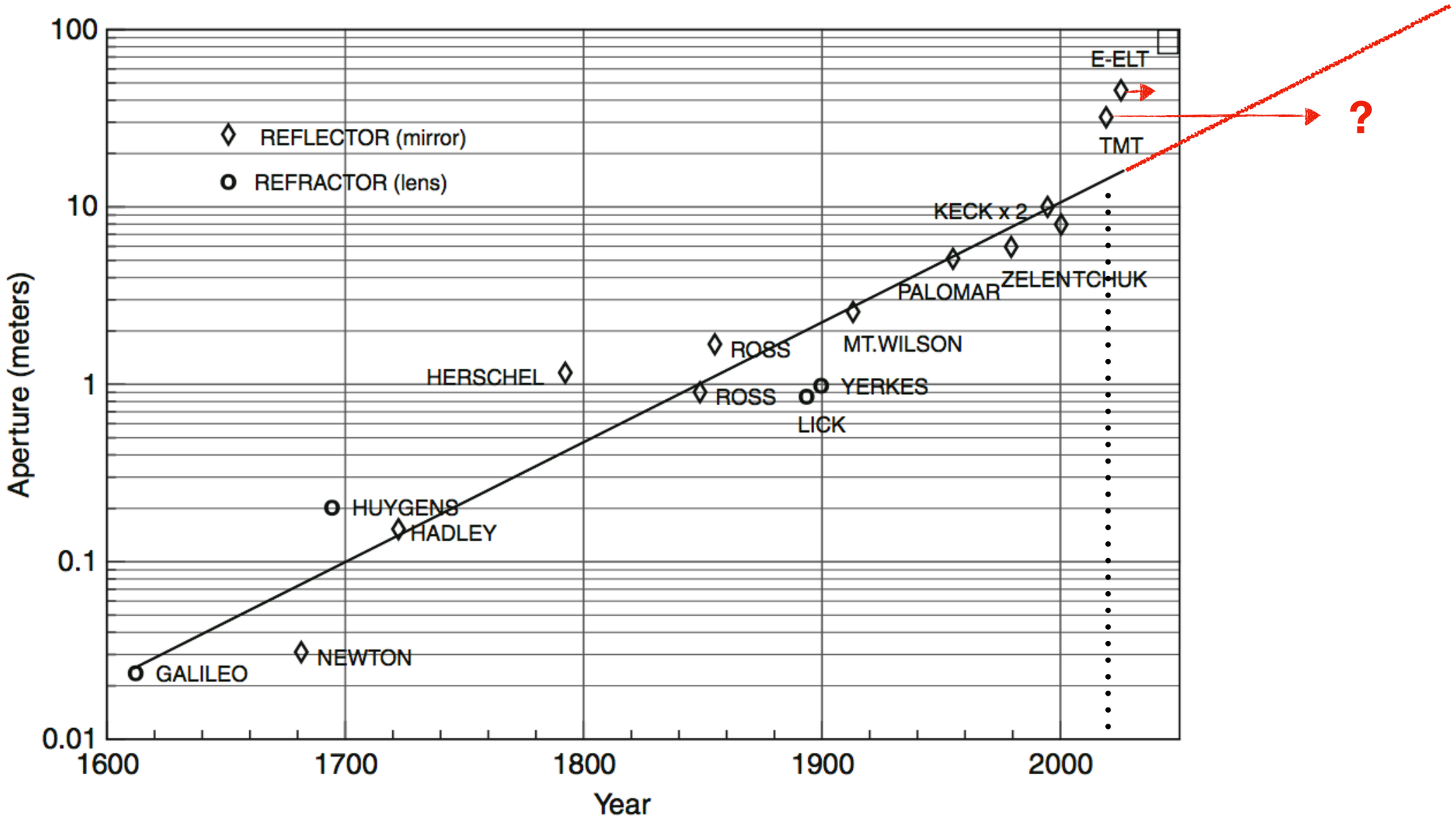


Fig. 1.8 Evolution in the diameters of astronomical instruments in the optical range. Note the gradual disappearance of refractors (○) in favour of reflectors (◇). Only the main stages have been shown here. The Keck I and II telescopes and the European Very Large Telescope (VLT) represent a whole family of instruments in the range of 8–10 m, not all shown here (see Sect. 5.2), that came into service around the year 2000. The Thirty Meter Telescope (TMT) in California and the European Extremely Large Telescope (E-ELT) are shown with projected completion dates. (Figure adapted by P. Bely)

Telescopes

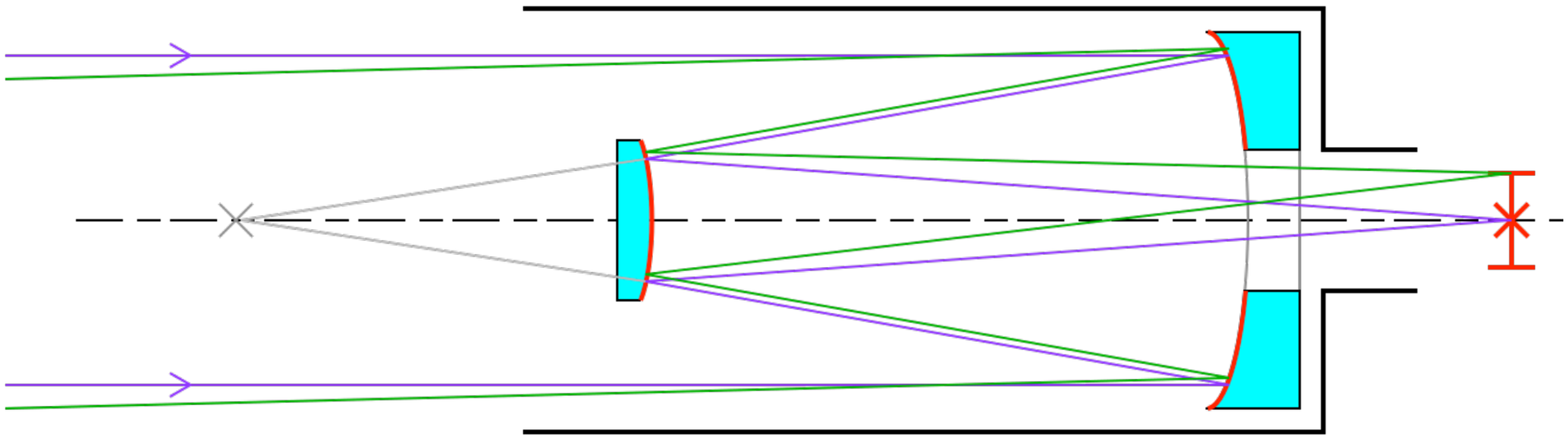
- Almost all professional astronomical telescopes are *reflectors. (Why?)*

- In contrast to undergrad geometric optics, the source is always at “infinity”, so images are formed at the focal plane.

- Essentially the telescope maps the off-axis *angle* of an incident beam into an off-axis *displacement* of the image in the focal plane.

- The ratio (plate scale) depends on the focal length.

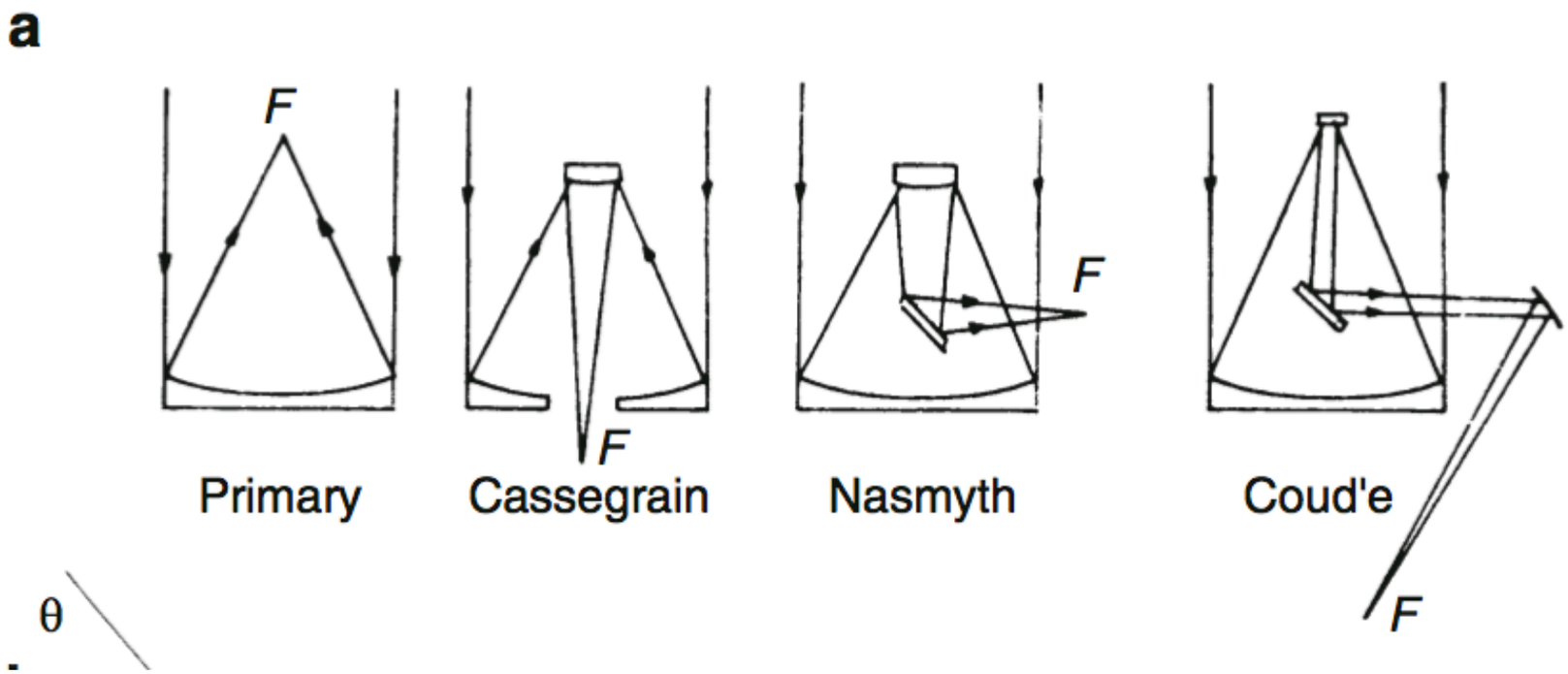
Telescope



Example: Cassegrain design

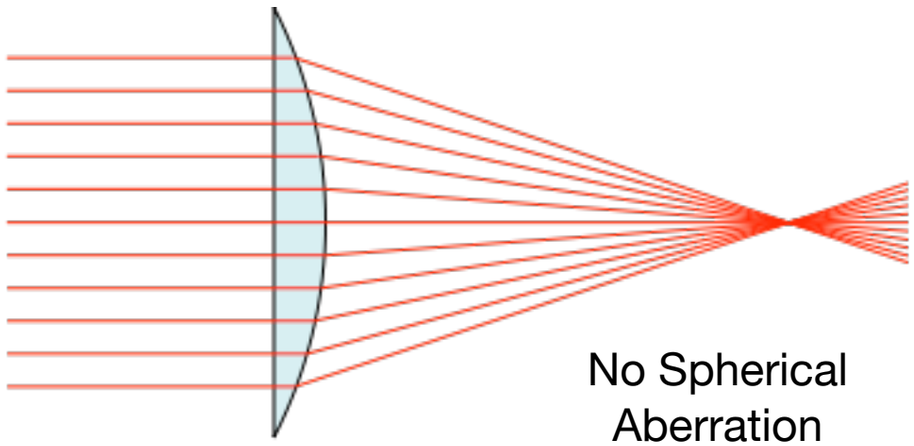
Credit: Krishnavedala - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=34884711

Telescope Designs

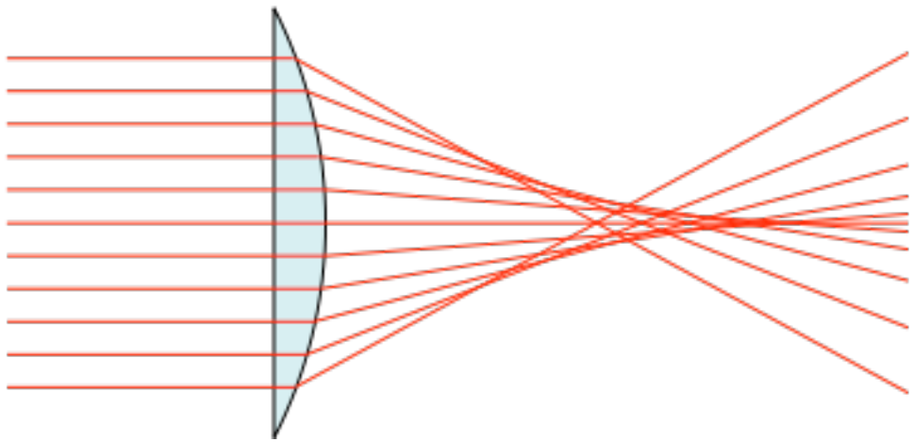


Many professional astronomical telescopes are a modification of Cassegrain called Richy-Chretien

Telescopes: Aberrations

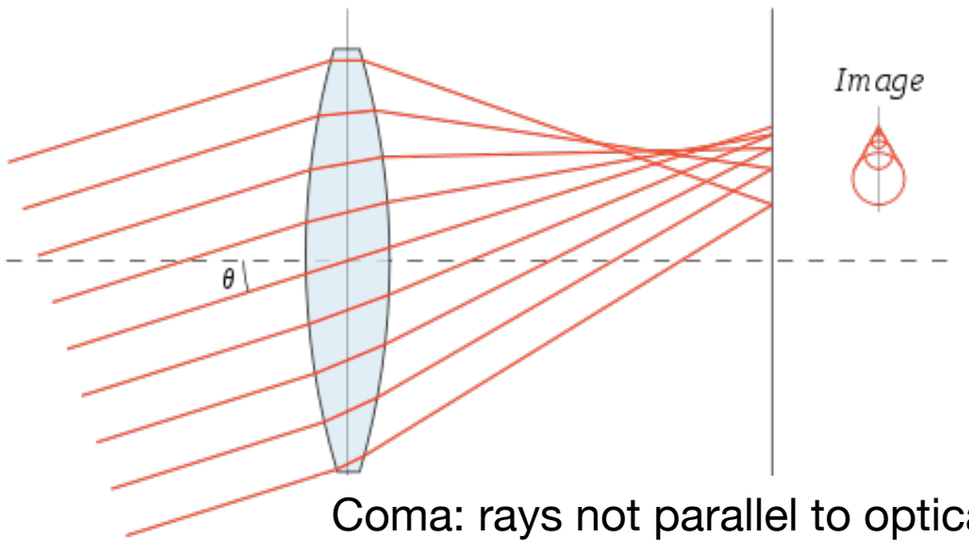


No Spherical Aberration



Spherical Aberration

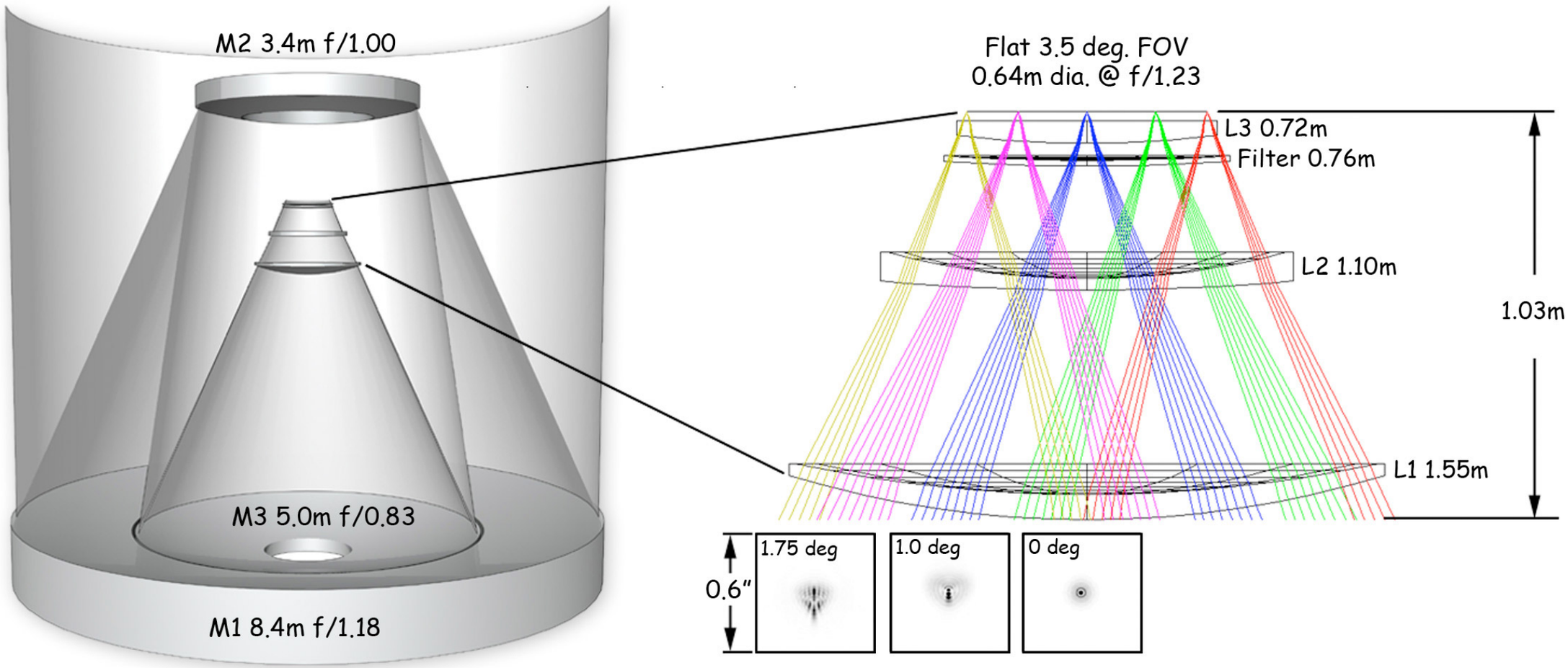
Spherical Aberration: rays near edge focussed differently



Coma: rays not parallel to optical axis

Credit: Mglg, uploaded to English Wikipedia, Public Domain, https://commons.wikimedia.org/w/index.php?curid=3655426

LSST telescope and camera



Telescope Characteristics

- Aperture: light collecting power D^2
  - Current state of the art for ground-based optical D = 8-10m.
  - Coming "soon" ELT D = 39.5 m
    - N.B. Hubble: 2.4m, JWST: 6.5m
- Field of View:
  - CFHT: 1 square degree, Rubin SST: 9.5 square degrees
- *Etendue*: Aperture x FoV
  - Relevant figure of merit for survey telescopes
- Focal length, *f*

Yerkes Observatory refractor (40″ lens at the same scale)
Williams Bay, Wisconsin, USA (1893)

Great Paris Exhibition Telescope
(49″ lens at the same scale)
Paris, France (1900)

Large Sky Area Multi-Object Fiber Spectroscopic Telescope (LAMOST)
Hebei, China (2009)

Gran Telescopio Canarias
La Palma, Canary Islands, Spain (2007)

Keck telescopes
Mauna Kea, Hawaii, USA (1993, 1996)

Thirty Meter Telescope
Mauna Kea, Hawaii, USA (planned)

Hooker Telescope (100″)
Mt Wilson, California, USA (1917)

Hale Telescope (200″)
Mt Palomar, California, USA (1948)

Hobby–Eberly Telescope
Davis Mountains, Texas, USA (1996)

Southern African Large Telescope
Sutherland, South Africa (2005)

Gemini North
Mauna Kea, Hawaii, USA (1999)

Subaru Telescope
Mauna Kea, Hawaii, USA (1999)

(1979–1998) (1999–)
Multiple Mirror Telescope
Mount Hopkins, Arizona, USA

Large Binocular Telescope
Mount Graham, Arizona, USA (2005)

Gemini South
Cerro Pachón, Chile (2000)

Vera C. Rubin Observatory
Cerro Pachón, Chile (2025)

BTA-6 (Large Altazimuth Telescope)
Zelenchuksky, Russia (1975)

Large Zenith Telescope
British Columbia, Canada (2003)

Gaia
Earth–Sun L2point (2014)

Kepler
Earth-trailing solar orbit (2009)

Very Large Telescope
Cerro Paranal, Chile (1998, 1999, 2000, 2000)

Extremely Large Telescope
Cerro Armazones, Chile (planned 2028)

James Webb Space Telescope
Earth–Sun L2point (2022)

Hubble Space Telescope
Low Earth orbit (1990)

Magellan Telescopes
Las Campanas, Chile (2000, 2002)

Giant Magellan Telescope
Las Campanas, Chile (planned 2030s)

Human at the same scale

0 5 10 m
0 10 20 30 ft

Overwhelmingly Large Telescope (cancelled)

Arecibo Observatory 305 m radio telescope at the same scale

FAST (Five-hundred-meter Aperture Spherical [radio] Telescope) at the same scale

Tennis court at the same scale

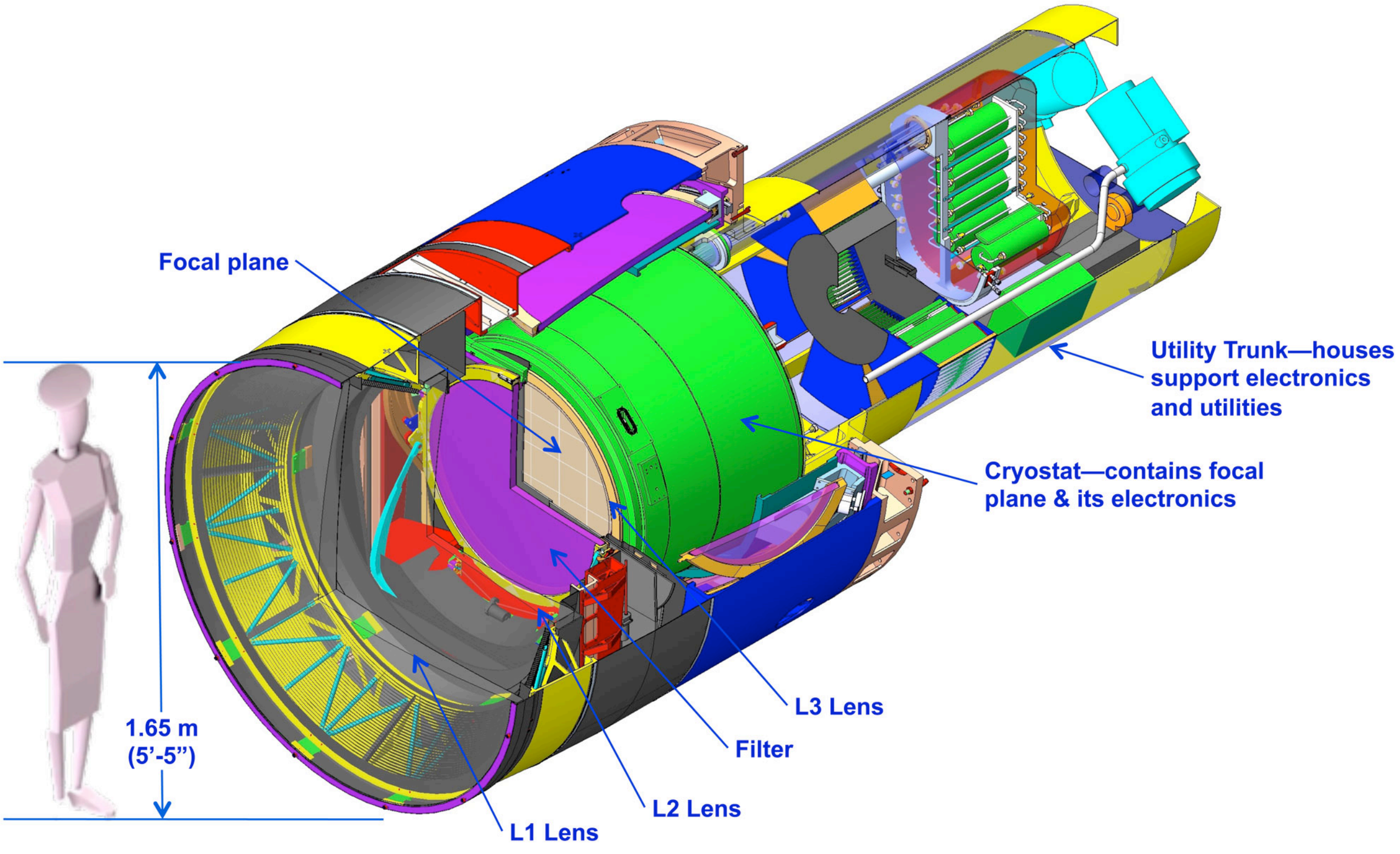
Basketball court at the same scale

Focal length and plate scale

- Focal length of the telescope determines the “plate scale”:
  - linear separation on detector from axis, $s = f\,\theta$
  - *plate scale* is the inverse, usually expressed in arcsec per mm = 206265” / *f* where *f* is in mm

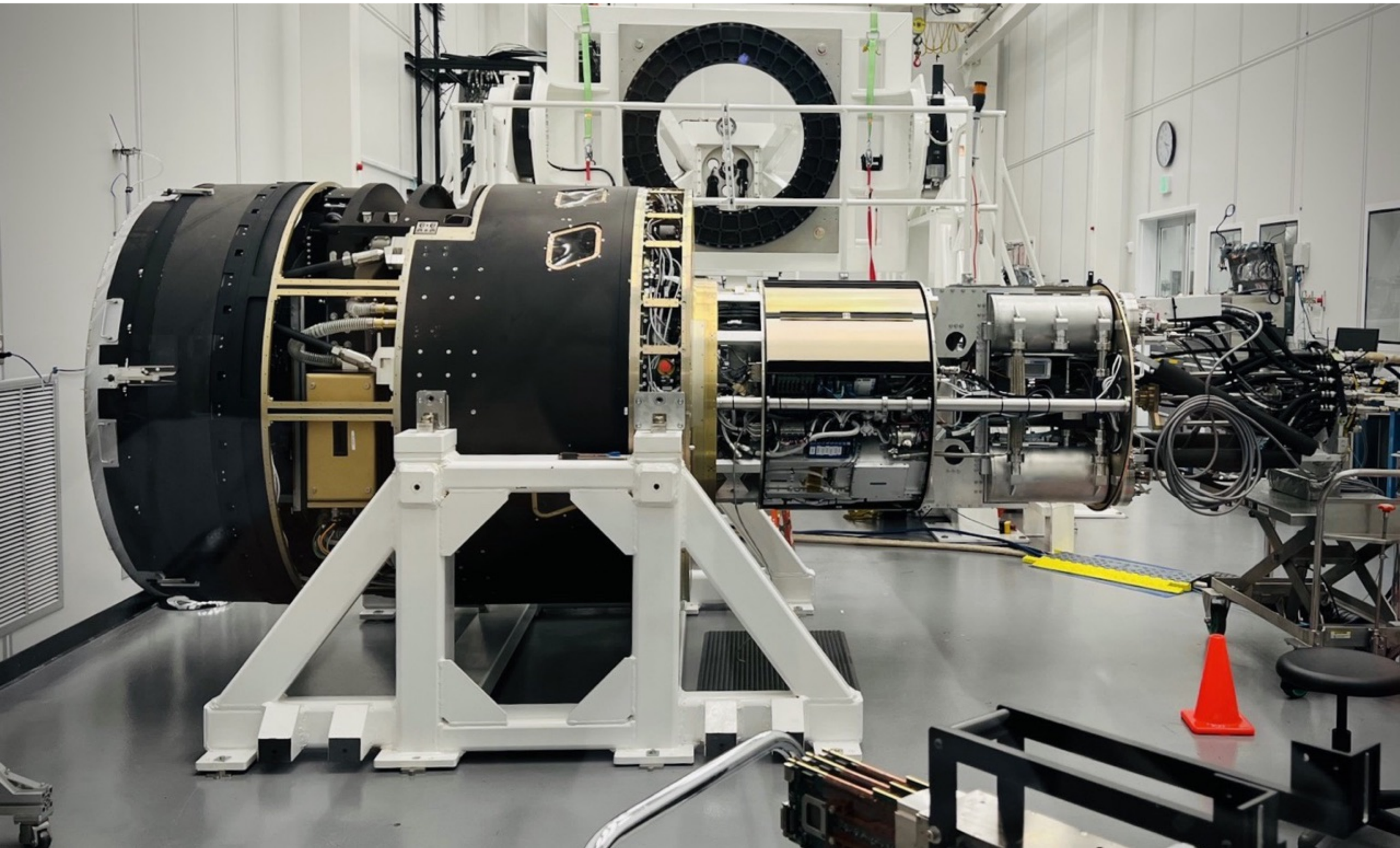
Example: Rubin Observatory Simonyi Survey Telescope

- f = 10.3m = 10300 mm, so this gives 20”/mm
- The focal plane is 640mm wide so this gives FoV 3.5 degrees across
- 3.2 Gigapixels made of 189 4k x 4k CCDs
    - 10 micron pixels -> 0.2 arcsecond sampling. The pixels *oversample* the expected PSF (0.7 arcsec). This is good.



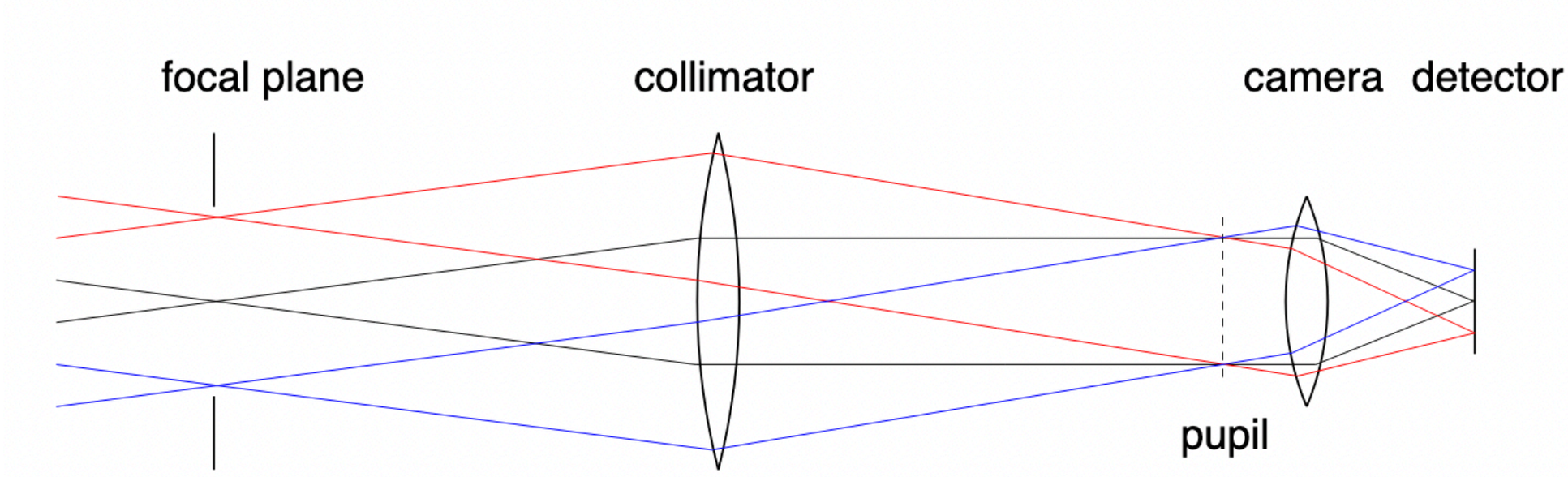
Camera ¾ Section

Example: Rubin Observatory Simonyi Survey Telescope



Reimaging optics

- Collimators and cameras allow a change of plate scale



From Weiner **Figure 3:** The (idealized) reimaging system with the paths of light from on-axis (black lines) and off-axis objects (red and blue lines).